



### Interacting Multipoles of Rank 3 and 3:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13}.\frac{1}{225}.$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225}.$$

[illegible]

## Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225}.$$

[illegible]

## Simplify Multipoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225} \cdot \left[ \begin{aligned} &-945 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{zz\gamma} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{zz\gamma} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{zz\gamma} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{zz\beta} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{zz\beta} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{zz\beta} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} \\ &-945 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\epsilon\epsilon} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\epsilon\epsilon} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\beta} \cdot \Omega_{z\epsilon\epsilon} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\gamma} \cdot \Omega_{z\beta\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\gamma} \cdot \Omega_{z\beta\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\beta} \cdot \Omega_{z\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\gamma} \cdot \Omega_{z\beta\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\gamma} \cdot \Omega_{z\beta\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\beta} \cdot \Omega_{z\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\gamma} \cdot \Omega_{z\beta\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\beta\gamma} \cdot \Omega_{z\beta\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\epsilon\epsilon} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\alpha} \cdot \Omega_{z\epsilon\epsilon} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\gamma} \cdot \Omega_{z\alpha\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\gamma} \cdot \Omega_{z\alpha\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\alpha} \cdot \Omega_{z\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\gamma} \cdot \Omega_{z\alpha\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\gamma} \cdot \Omega_{z\alpha\gamma} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\alpha} \cdot \Omega_{z\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\alpha} \cdot \Omega_{z\delta\delta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &+105 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &+105 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{\alpha\alpha\gamma} \end{aligned} \right] \quad (4a)$$

## Exploit Traceless Conditions:

[illegible]

## Reduce Indices:

[illegible]

Add up:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225} \cdot \left[ \begin{array}{l} -2835 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{zz\gamma} \\ -2835 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{zz\beta} \\ -2835 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ +630 \cdot R_z^6 \cdot \Omega_{z\alpha\delta} \cdot \Omega_{z\alpha\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\gamma\gamma} \\ +630 \cdot R_z^6 \cdot \Omega_{z\beta\delta} \cdot \Omega_{z\beta\delta} \\ +630 \cdot R_z^6 \cdot \Omega_{z\gamma\delta} \cdot \Omega_{z\gamma\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\delta} \cdot \Omega_{\beta\beta\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\delta} \cdot \Omega_{\gamma\gamma\delta} \\ +315 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\delta\delta} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\gamma} \end{array} \right] \quad (7)$$



Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225} \cdot \left[ \begin{aligned} &-2835 \cdot R_z^6 \cdot \Omega_{xxx} \cdot \Omega_{xxx} \\ &-2835 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} \\ &-2835 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \\ &-2835 \cdot R_z^6 \cdot \Omega_{xxx} \cdot \Omega_{xxx} \\ &-2835 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} \\ &-2835 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \\ &-2835 \cdot R_z^6 \cdot \Omega_{xxx} \cdot \Omega_{xxx} \\ &-2835 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} \\ &-2835 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \\ &+315 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ &+315 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ &+630 \cdot R_z^6 \cdot \Omega_{xxz} \cdot \Omega_{xxz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \\ &+315 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &+630 \cdot R_z^6 \cdot \Omega_{xxz} \cdot \Omega_{xxz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xxz} \cdot \Omega_{xxz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &+630 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \\ &+315 \cdot R_z^6 \cdot \Omega_{zz\delta} \cdot \Omega_{\beta\beta\delta} \\ &+315 \cdot R_z^6 \cdot \Omega_{zz\delta} \cdot \Omega_{\gamma\gamma\delta} \\ &+315 \cdot R_z^6 \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\delta\delta} \\ &-45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ &-45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &-45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ &-90 \cdot R_z^6 \cdot \Omega_{xxx} \cdot \Omega_{xxx} \end{aligned} \right] \quad (8a)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = \begin{bmatrix} -\frac{111}{5} \cdot R_z^{-7} \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ +\frac{7}{5} \cdot R_z^{-7} \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ +\frac{7}{5} \cdot R_z^{-7} \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ +\frac{7}{5} \cdot R_z^{-7} \cdot \Omega_{zz\beta} \cdot \Omega_{\beta\gamma\gamma} \\ +\frac{7}{5} \cdot R_z^{-7} \cdot \Omega_{zz\delta} \cdot \Omega_{\beta\beta\delta} \\ +\frac{7}{5} \cdot R_z^{-7} \cdot \Omega_{zz\delta} \cdot \Omega_{\gamma\gamma\delta} \\ +\frac{7}{5} \cdot R_z^{-7} \cdot \Omega_{zz\gamma} \cdot \Omega_{\gamma\delta\delta} \\ -\frac{1}{5} \cdot R_z^{-7} \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ -\frac{1}{5} \cdot R_z^{-7} \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -\frac{1}{5} \cdot R_z^{-7} \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -\frac{6}{5} \cdot R_z^{-7} \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\ -\frac{2}{5} \cdot R_z^{-7} \cdot \Omega_{yyy} \cdot \Omega_{yyy} \end{bmatrix} \quad (9)$$